

Ginger Schmidt

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EDUCATION

Harvard-MIT Health Sciences and Technology

Cambridge, MA

Ph.D. Medical Engineering and Medical Physics, MIT Electrical Engineering concentration

May 2027

Thesis: *Asynchronous optical coherence elastography for minimally invasive ophthalmic and intravascular imaging.*

- MIT Coursework: Optics & Photonics, Nonlinear Optics, Biomedical Signal & Image Processing, Advances in Computer Vision (Neural Networks), Digital Control Systems
- Harvard Medical School Coursework: Cardiovascular Pathophysiology, Respiratory Pathophysiology, Genetics, Immunology, Human Pathology

Harvey Mudd College

Claremont, CA

B.S. Engineering

May 2021

- Advanced Systems & Signal Engineering, Dynamics of Elastic Systems, Differential Equations, Linear Algebra, Data Structures (C++), Digital & Analog Electronics, Robotics State Estimation, Robotics Motion Planning

RESEARCH EXPERIENCE

Massachusetts General Hospital and Harvard Medical School

Boston, MA

Graduate student researcher

June 2021 – present

- Developing signal processing methods to extract quantitative tissue dynamics from OCT phase measurements and enabling multi-frequency, 3D mechanical characterization.
- Designing miniaturized micromotor-based imaging catheter for ultrafast coronary artery imaging.
- Analytical and computational inverse methods for solving wave-propagation problems in isotropic and transversely isotropic media to recover tissue mechanical properties from measured displacement fields.
- Advised by Professors Néstor Uribe-Patarroyo and Brett E. Bouma

Lab for Autonomous and Intelligent Robotics

Claremont, CA

Undergraduate student researcher

Sept 2019 – May 2021

- Designed centimeter-scale autonomous underwater glider that conducts electrolysis of water to modulate its center of mass, performed controls and stability simulations, validated via autonomous depth tracking.
- Advised by Professors Christopher Clark and Matthew Spencer

INDUSTRY EXPERIENCE

Johnson & Johnson, Robotics and Digital Surgery

Santa Clara, CA

Mechanical Engineering & Robotics Intern

August 2020 – May 2021

- Characterized magnetic crosstalk interference in neighboring encoders.
- Designed 6-axis motor fixture with torque sensors and encoders for surgical instrument manipulator verification & validation testing. Selected electrical and control system components for the fixture

Raytheon Technologies

El Segundo, CA

Engineering Intern, Space and Airborne Systems

May 2019 – August 2019

- Wrote product specifications for thermal vacuum chamber for level 5 space hardware testing.
- Designed test fixture and test procedure for wire marking durability analysis, performed FEA in CREO.

Auris Health

Redwood City, CA

Mechanical Engineering & Robotics Intern, Advanced Development

May 2018 – January 2019

- Designed electro-mechanical packaging solution for a surgical instrument manipulator to fit torque sensors in a constrained volume while improving DFM and DFA.
- Mechanical and electrical assemblies of surgical robotic arms with brushless motors, sensors, slip rings, and harmonic drives.

FELLOWSHIPS

2026-27	MIT Next-Generation Medical Imaging Graduate Fellowship
2025-26	MIT Health and Life Sciences Collaborative Graduate Fellowship (HEALS)
2022-25	National Science Foundation Graduate Research Fellowship (NSF-GRFP)
2021-22	MIT Presidential Fellowship
2019-21	Jude and Eileen Laspa Fellowship in Autonomous Systems
2017-21	Harvey S. Mudd Merit Scholarship
2017	National Merit Scholarship

AWARDS

2025	Best Presentation (International Tissue Elasticity Conference)
2025	Best Poster Award (SPIE Photonics West, Elastography & Tissue Mechanics)
2024	Best Poster Award (Gordon Research Conference, Optics in Medicine)
2023	Best Poster Award (Biophotonics Summer School)
2022	50 Women in Robotics You Need to Know About (Robohub)
2021	Johnson Excellence in Engineering Award (Harvey Mudd College)
2020	Alford-Gilkeson Clinic Award (Harvey Mudd College)

SUPERVISING & MENTORSHIP

Isabel Sperandio '25	B.S. Mechanical Engineering, MIT. Ph.D. Student in Mechanical Engineering at MIT, continuing in our lab.
Huda Abdelghani '27	B.S. Mechanical Engineering, MIT. Best Poster at the New England Symposium for Biomedical Optics (2025).
Eric Zou '29	B.S. Electrical Engineering, MIT.

TEACHING

Spring 2021	HMC E206: Robotics State Estimation, TA
Spring 2021	HMC E102: Advanced Signals & System Engineering 2, Head Grader and Tutor
Fall 2020	HMC E205: Robotics State Estimation, TA
Fall 2020	HMC E101: Advanced Signals & System Engineering 1, Head Grader and Tutor
Fall 2020	HMC E79: Introduction to Systems Engineering, TA
Spring 2020	HMC E80: Experimental Engineering, Lab Proctor
Fall 2019	HMC E79: Introduction to Systems Engineering, TA

SERVICE

2025-	MIT Next-Generation Medical Imaging committee member, organizing institute-wide monthly seminars
2025-	Graduate Student Advisory Group for Engineering: Advisor to the Dean of the MIT School of Engineering
2025-	HST Joint Council, MD Curriculum Committee Member
2025-26	Graduate Admissions Interviewer for the Harvard-MIT Health Sciences & Technology Ph.D. program
2021-24	MIT Application Assistance Program: Mentored underrepresented students in grad school applications.
2018-21	Prison Education Project: President and instructor, computer science and physics.
2019-21	Resident Assistant and Dorm Proctor at Harvey Mudd College

EXTRACURRICULAR PROJECTS

Battlebots

2017 – present

Co-captain

- Season 6 World Championship Winner of Battlebots on the Discovery channel, appearing on Seasons 3-7.

Pressona

2025 – present

Founding engineer

- Our core technology is a closed-loop “autopilot” for arterial blood pressure management in the OR and ICU which combines real-time signal processing with adaptive control to help clinicians keep patients in a safe thermodynamic window while reducing the cognitive burden of constant titration.
- In collaboration with CEO Taylor Baum, based on her Ph.D. thesis work at MIT with Professor Emory Brown.

PUBLICATIONS

1. **Ginger Schmidt**, Ryan McAuley, Brett E. Bouma, and Néstor Uribe-Patarroyo, "Multi-frequency asynchronous optical coherence elastography by direct demodulation," Biomedical Optics Express 17, 4947-4967 (2026)
2. **Ginger Schmidt**, Ryan McAuley, Brett E. Bouma, and Néstor Uribe-Patarroyo, "Asynchronous optical coherence elastography and directional phase gradient analysis," Journal of Biomedical Optics (2025).
3. Manisha Singh, Julia Situ, Carly Long, Diego Quevedo-Moreno, **Ginger Schmidt**, Néstor Uribe-Patarroyo and Ellen T. Roche, "Coronary Artery Flow Modeling in an Ex-vivo Biorobotic Heart," 2025 IEEE 8th International Conference on Soft Robotics (2025).
4. **Ginger Schmidt**, Brett E. Bouma, and Néstor Uribe-Patarroyo, "Asynchronous, semi-reverberant elastography," Optica 11, 1285-1294 (2024).

INVITED TALKS

1. Optica Biophotonics Congress in Fort Lauderdale, Florida: "Shear wave elastography through a helically-scanned catheter using asynchronous optical coherence elastography." (2026)
2. SPIE Photonics West in San Francisco, CA: "Towards intraluminal elasticity measurements via an endoscope with asynchronous optical coherence elastography." (2026)
3. MIT Medical Imaging Seminar Series in Cambridge, MA (2026)
4. New England Symposium for Biomedical Optics (NESBO) in Cambridge, MA (2025)
5. Wellman-University of Tokyo Graduate Student Symposium in Tokyo, Japan (2025)
6. Drexel University ECE-101 Guest Lecturer in Philadelphia, PA (2023)

CONTRIBUTED TALKS

1. SPIE Photonics West in San Francisco, CA: "Non-destructive mechanical phenotyping of 3D cartilage tissue models for osteoarthritis with asynchronous optical coherence elastography." (2026)
2. SPIE Photonics West in San Francisco, CA: "Biomechanical assessment of coronary arteries under recapitulated physiological motion with asynchronous shear wave optical coherence elastography." (2026)
3. International Tissue Elasticity Conference in Seattle, WA: "Elasticity measurement of the human cornea in vivo with 3D asynchronous optical coherence elastography." (2025)
4. International Tissue Elasticity Conference in Lyon, France: "Asynchronous semi-reverberant elastography." (2024)
5. SPIE Photonics West in San Francisco, CA: "Overcoming the spatial and temporal coherence limitations of reverberant elastography in raster-scanned OCT systems." (2023)
6. SPIE Photonics West in San Francisco, CA: "Enabling quantitative shear wave elastography in conventional optical coherence tomography in vivo." (2023)